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A Practical Approach to AI-Based Route Optimization

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Executive Summary

This whitepaper explores a practical approach on utilization of artificial intelligence (AI) to perform vessel route optimization in offshore oil and gas logistics. It provides a pragmatic, data-centric perspective on how a Management Software-as-a-Service (SaaS) platform combined with human business knowledge and interaction can be combined to achieve significant gains in logistics performance. Drawing from experience from real world implementations with major industry operators, this document highlights the key takeaway that the success of AI solutions is less dependent on algorithmic sophistication and more reliant on data input, how it is structured and contextualization. Through a multi-phased implementation methodology, simplification of route optimization goals and a practical step by step data gathering process, operator companies are able to reduce vessel usage, improve scheduling efficiency, and adapt to operational constraints with measurable success.

Problem Statement

When utilizing vessels to deliver equipment to offshore locations, a sailing route must be defined, taking various information into account, such as but not limited to: when equipment must be on site, load capacity of the vessels, cost of equipment on board, opening hours on rigs, delivery timelines on supply based, weather conditions and other operational constraints.

When the cargo and vessels to be served increases, this job becomes impossible to perform optimally for a human, of which a computer might be better suited to assist in finding scenarios on how to meet the offshore demand whilst focusing on the most cost-efficient vessel schedule.

This applies to both setting up fixed route schedules, but also how to define routes on an ad hoc basis or how to solve sudden challenges such as operational disruptions requiring immediate re-prioritization of deliveries or sailing routes.

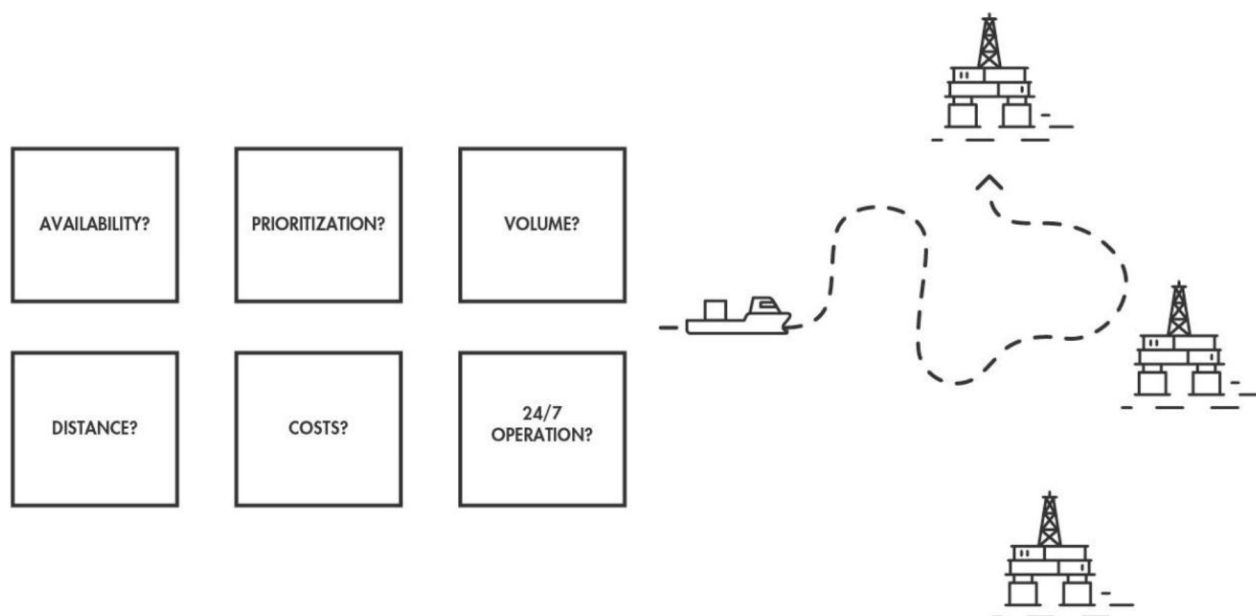


Figure 1—The concept of utilizing multiple parameters to define a schedule for route optimization

Utilizing AI to optimize route scheduling

The concept of utilizing AI to optimize route scheduling is based on the concept of inputting relevant data to an AI based optimizer tool which can process multiple possible scenarios and optimize the routes to try and reach the best total rating of a proposed route. This tool can assist a human in making schedule decisions as a part of a normal planning process, and over time maybe automatically and continuously assign routes to vessels and re-schedule when required.

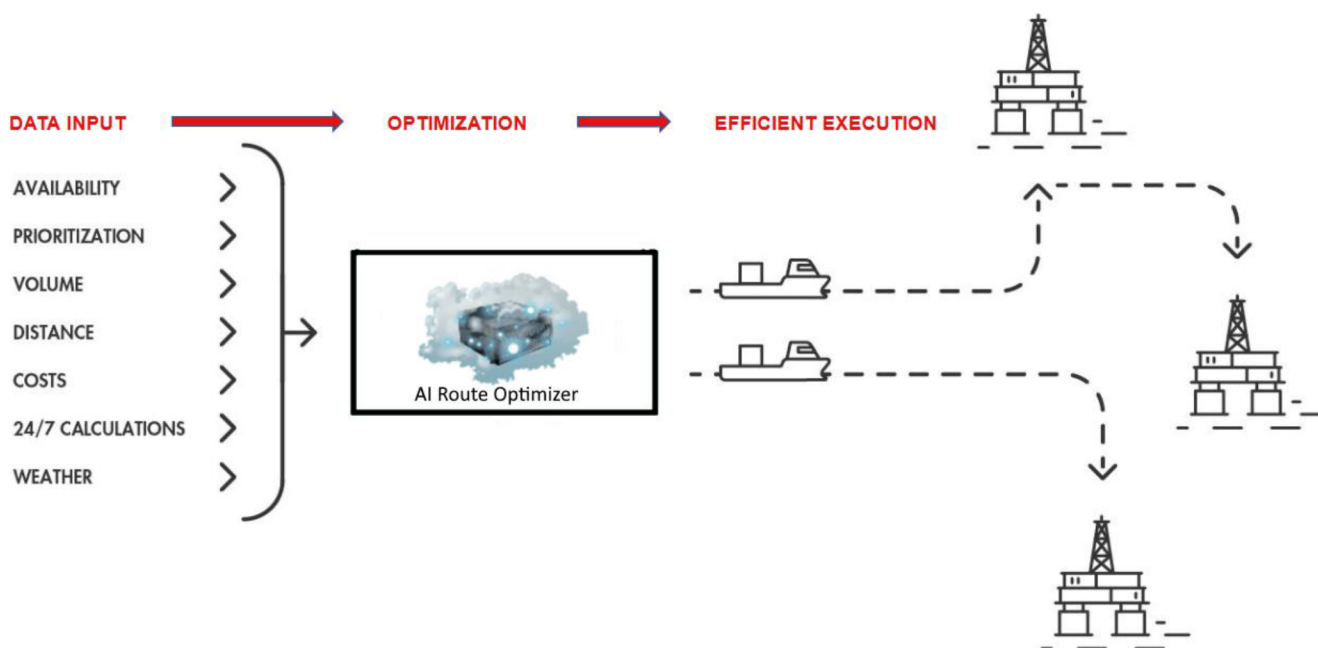


Figure 2—An AI-based route optimizer model performing automated scheduling

The Importance of Data Input

The success of any AI tool usage, whether in logistics or other business areas is reliant on the quality, relevance, and structure of the data it is served. AI models, particularly modern-age large language models

(LLMs), are highly sensitive to how questions are phrased and the context provided with it, and also the sequencing of it. Put simply the exactly same model can produce very different responses depending on how a question is framed. As an example, asking an LLM model "What's the best way to reduce the operational costs?" may provide a very generic suggestion, while a question with more context like "Our current OSV fleet is of size X, is utilized like 50% and fuel for travel are x amount per nm, please identify the top three cost-saving opportunities for our logistics operations?" can result in far more tangible and actionable responses. Similarly, machine learning AI models that are trained on poorly labeled or partially deviant data may misinterpret patterns. This again can lead to unreliable predictions. In an executive decision making world, this means that even the most advanced AI tools are only as effective as the clarity and completeness of the input they get. Without a well established data foundation and being provided with structured context, AI models can amplify data ambiguity rather than the opposite.

The role of a management SaaS

To fully realize the potential of AI, an organization must go further than just collecting data, it must be live, complete and organized properly. This is where a management SaaS solution comes into play. By being able to integrate information from all stakeholders across the logistics chain: Suppliers, transport providers, operational planners, external vendors, warehouse managers, field operators; a SaaS platform creates a single source of truth in real time that AI models can interpret. It ensures that the data is not only captured in real time but also includes the needed metadata. This includes time constraints, resource availability and cost, and environmental conditions. Without this as a foundation, AI models are left to try to infer meaning from fragmented or inconsistent information, often leading to suboptimal or misleading output. For example, an AI model might recommend a route that appears efficient on paper but fails to account for a maintenance window or a regulatory constraint that wasn't captured in the data. An AI model might also attempt to finish delivering equipment as soon as possible when the lack of relevant volumes ahead of time is missing. A centralized SaaS platform can help mitigate this by standardizing the data formats, enforcing data quality, and embedding business logic to ensuring that the AI models are always working with the correct information at the right time. This means that the SaaS layer is both the enabler and the safeguard for AI-driven decision support.

Potential Savings

The strategic implementation of AI-powered route optimization, supported by a centralized management SaaS platform has the potential to unlock substantial value across offshore logistics operations. For a fleet of 10 offshore supply vessels (OSVs), which are among the most cost-intensive assets in the industry, even a conservative 20–30% reduction in vessel usage can translate into millions in annual savings. Assuming an average daily operating cost of \$25,000 per vessel, a 25% reduction in utilization could save approximately more than \$22 million annually. Beyond the financial impact, the environmental benefits are equally compelling. Reducing fleet activity by 25% could equate to removing the emissions of 30,000 cars, 2 million NO_x-equivalent vehicles, and 6 million SO_x-equivalent vehicles from the equation. [1] These reductions contribute directly to operator companies bold ESG goals, and ensures regulatory compliance. Fewer vessel movements and cargo lifts also mean reduced exposure to HSE risks—fewer personnel involved in operations, less time in hazardous environments, and thus a lower chance of incidents. Moreover, by automating route planning and decision-making through AI, organizations can reduce the possibilities for human error, which remains a leading cause of inefficiencies and safety incidents.

The lack of current success

While the potential savings from AI-driven optimization are substantial and seems well know, many real-world attempts to operationalize these solutions have fallen short. [3]

Real world experience has shown that adoption has been limited, and the key findings of this is the following:

- Lack of proper usage of all relevant modules for data capture in management SaaS for all stakeholders
- Lack of timely data to process optimization (specifically Cargo demand planned days ahead of mobilization for both Outbound and especially Inbound volume)
- The optimizer has not implemented a feedback process to change the model based on end user input.

A practical approach to implementation of AI models

To successfully deploy AI tools to support route optimization for the offshore supply vessel operations, a methodical approach is required. Such an approach can be compared to the current gradual evolution of self-driving cars. While the end goal of a full autonomy self-driving world is very compelling, experience has shown that revolutionary leaps often fall short without a gradual foundational readiness. Incremental, well-structured implementations seem to be the goal that leads to sustainable success.

Below a practical stepwise approach on implementation of AI models for route optimization within the Oil and Gas industry is outlined.

Step 1: Implement a management SaaS as a unified data foundation

Just as self-driving vehicles rely on correct maps and a large set of real-time sensor information, the offshore logistics domain must begin with a centralized, structured and live data environment. This includes a means to continuously record and review vessel schedules, cargo priorities, weather forecasts, port availability, and operational constraints, ensuring consistency and accessibility when required.

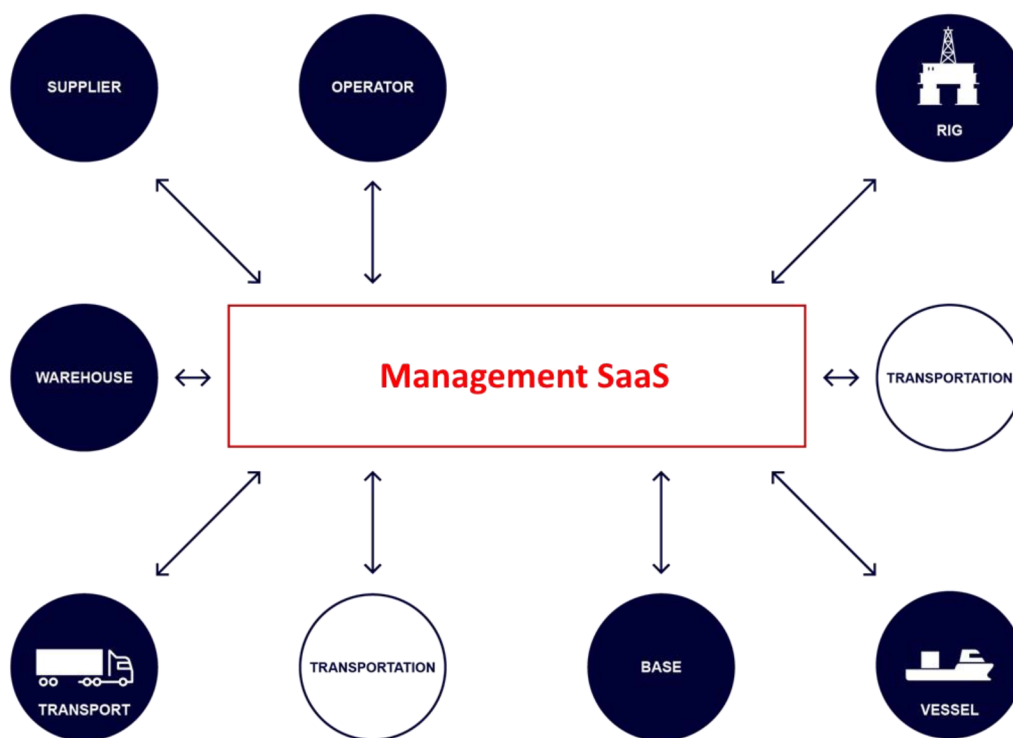


Figure 3—Utilizing a management SaaS as a communication and collaboration platform for data gathering

Step 2: Define Clear Operational Objectives

Before deploying AI, it's critical to define what the result of a good optimization is. An objective score metric of an suggested route will be different depended of whether the goal is to reduce fuel costs, minimize fuel usage, deliver rental equipment as close to required on site time, or a balance of all three? Achieving clear objectives score models ensure alignment with business requirements.

Therefore, each route a vessel optimizer creates must result in a corresponding score level, and it is important to define how such a score is measured in order to make the AI tool optimize towards the desired result. These objectives must be defined by humans and translated into a metric that can be used to automatically classify the rating of a scenario. Different metrics that can be emphasized is e.g trying to use as less fuel as possible, trying to reduce the number of vessels operating at the same time, trying to minimize the total number of rental days, trying to deliver the cargo as close as possible to the ROS date.

Step 3: Start as Assistant

Rather than aiming for full automation with no human interaction and a complete and final AI model from the start, instead begin with achieving AI-assisted decision making. This is a mirror on how the car self-driving technology started with lane assist and adaptive cruise control.

Step 4: Validate in Real-World and Real-Time

AI results should be tested against historical data and ideally compared in parallel during live operations. This step helps identify gaps in data and logic before full deployment, and provides the option to modify the model as it evolves.

Step 5: Iterate and Scale Usage

As confidence in the AI model grows, expand the usage. Move from assisted planning to more autonomous (but not complete) routing. Each phase should include feedback loops to refine the model and improve it.

Step 6: Monitor and Evolve

AI models are not static. By implement a process of continuous monitoring, performance audits, KPI reviews and stakeholder feedback back to the model the system can adapt to changing operational realities.

Just as self-driving cars didn't (and still haven't) revolutionized transportation from one day to another, the utilization of AI in offshore logistics must also evolve through structured steps. The key is not to rush the implementation, but to build it on a foundation of trust, transparency, and operational relevance, continuously evolving.

Data Gathering and user interaction using a Management SaaS

To achieve valuable results from AI-based route optimization, the quality and structure of input data are paramount. It is crucial to understand how the data is connected, which data to gather, which information to ignore, and how to prepare it in a structured manner for the software engine to utilize. Interaction with end users who have deep knowledge of the business domain is also essential to define success criteria on a case-by-case basis.

Many factors contribute to creating an acceptable route schedule, and not all of them can be defined technically before gathering extensive experience data, or sometimes not at all. For example, certain vessels may not be able to visit specific rigs due to technical or legal requirements, incompatibility of technical equipment at specific offshore sites, unforeseen events, unknown constraints, or even personal preferences.

Both before running an optimization and when reviewing its results, human oversight is crucial. This oversight allows for the identification and correction of any inaccuracies in the data. By making these human alterations, stakeholders can ensure that the data fed into the AI system is accurate and reflective of

real-world conditions. Additionally, a good AI model requires human feedback to properly train itself and provide more valuable results over time.

A technical AI optimization tool is just a small part of a connected workflow and will not provide valuable results before being implemented in a connected process between several stakeholders all connecting onto the workflow.

Visual Route Planner Interaction

Business users of the Management SaaS of which AI route optimization support is implemented require a visual tool to properly review data prior to AI optimization and to fill in missing data. After an optimization is run, the end user must be able to visually fetch, display and prepare results of optimization and provide feedback score must be in place. This will ensure the possibility for real-world adoption of AI optimization and the means to understand shortcomings and possibilities in enhancements of the data provided or the AI models in use.

Vessel Information

To enable the automatic selection of offshore supply vessels to perform cargo delivery, an up-to-date master dataset on each vessel is essential. This includes information about the vessel's current position and current future sailing plan, the cargo that is already loaded on board, both on top of the deck and inside the bulk tanks below, along with its destination and delivery instructions. Additional operational parameters such as vessel current fuel levels, the upcoming crew change schedule, and the properties of technical equipment should also be captured. This ensures that a vessel can be fit for an intended voyage. Cargo carrying units (CCUs) that have been marked for storage onboard or roundtrip back to its original destination must also be identified, as they impact available deck space and influence future transport capacity.

A management SaaS platform is responsible for gathering, maintaining, and operationalizing this data. By integrating real-time vessel data, cargo manifest content and operational schedules into a single live and digital platform, the AI model has access to the context required for smart vessel assignment. Without this detailed information, an optimizer might assign vessel that is not compatible.

Cargo Requirements

The end goal of optimization of offshore supply vessel routes is at its core driven solely by the cargo to be delivered, and their corresponding required on-site (ROS) delivery date. For rental equipment, delivering as close as the required date (and also returning as soon as possible thereafter) will significantly reduce the operational cost. To enable efficient optimization, this data must be input timely and as complete as possible by all the suppliers delivering it. A management SaaS platform facilitates this by acting as a centralized interface for suppliers, logistics planners, and marine operators to submit and update operational data in real time. The earlier the cargo requirements are captured, the more accurate and complete the AI model has in generating optimal routing patterns. By ensuring early and consistent data input through a SaaS platform, organizations enable proactive planning and keep last-minute changes to a minimum.

Work Plans

To effectively manage offshore logistics and enable true dynamic route changes, continuous visibility into the required delivery and backload times for equipment and cargo is required. Ideally, all cargo to be transported should be linked to a corresponding Work Order plan or drilling timeplanner section, which provides information on when specific materials are needed on site and when they are expected to be returned. This connection connects operations with logistics planning, allowing all roles to make informed decisions based on the latest operational requirements. A management SaaS platform facilitates this linkage by integrating planning data with cargo deliveries in real time. As operational timelines shift, e.g due to weather shutdowns, drilling progress, unforeseen changes or operational decisions, the SaaS platform can

ensure that most updated ROS (Required on Site) dates are always reflected. Without this integration, planners risk operating on outdated assumptions. By embedding cargo into the broad planning framework through a centralized SaaS solution, organizations can achieve a more agile, responsive, and cost-effective logistics operation.

Weather Forecasts

Offshore operations can be sensitive to weather conditions. By integrating a management SaaS with a weather forecast services through API connections, the tool can automatically flag weather event windows that may restrict offshore deliveries. Such real-time insight allows the AI model to dynamically adjust routing plans to avoid disruptions. This capability is especially critical in regions with volatile weather patterns, where timely adjustments can avoid costly downtime.

Rig Blocking Events

Events such as helicopter visits, crew transfers, maintenance windows, or other operational activities can restrict when a vessel is allowed to arrive at or depart from a rig. These constraints must be accounted for in the route optimization process. To ensure that AI-generated plans are both realistic and executable, this type of scheduling information must be input and confirmed by end users in a centralized management SaaS platform. By capturing these constraints in real time and integrating them into the operational dataset, the platform ensures that the AI model has a complete and accurate picture of logistical limitations. Without this input, an AI optimizer may propose routes that will conflict with these critical activities and subsequently lead to delays and even safety risks.

Rental Cost

If reducing the cost is the goal of optimization of vessel routes, minimizing the rental days is the key objective. To achieve this, route optimization must be closely aligned with both financial information. Delivering rental equipment as close as possible to its actual required on site date not only reduces unnecessary rental costs but might also improve equipment utilization across assets. To support this, requiring supplier companies to input detailed equipment cost usage within a management SaaS platform can be leveraged to compare and prioritize equipment based on their actual rental cost profiles. This allows an AI model to prioritize which items should be delivered in which sequence and which can be consolidated; based on financial impact rather than just operational convenience.

This cost-based prioritization can be coupled with real-time operational planning data. By linking each cargo item to a Work Order plan or drilling time planner as described above, the SaaS system ensures that the Required On Site (ROS) dates are always updated. This enables the AI optimizer to adjust delivery schedules in response to changes in drilling progress and operational work order plans, equipment readiness, or operational delays. Without this connection, the optimizer may prioritize based on outdated or incorrect assumptions, leading to unnecessary costs.

A centralized SaaS platform facilitates this entire process by acting as the bridge between financial data, operational planning, and logistics execution.

Other relevant Master Data

An enabler of successful AI-based route optimization in offshore logistics is also the availability of comprehensive and accurate master data. Such a foundational layer includes a wide range of parameters such as vessel technical specifications (e.g., deck space, tank capacity, crane capabilities), rig opening hours, base working schedules, overtime work compensation rules, rig downtime costs, cargo handling constraints at supply bases, and transportation lead times from vendors to the supply base. Each of these data points plays a role in shaping the feasibility and efficiency of routing decisions. For instance, a vessel may be technically capable of carrying a load, but if the rig is closed during the planned delivery window or if the

base cannot handle the cargo type, the route becomes invalid. Similarly, to understand the cost implications of delays or overtime allows the AI model to perform trade-offs and prioritize to minimize financial impact.

A management SaaS platform can maintain and operationalize such master data. It serves as a centralized repository where stakeholders are able to review and update this information in real time. By embedding master data into the logistics ecosystem through a SaaS platform, organizations lay the groundwork intelligent offshore logistics planning.

AI result process and manual feedback to engine

To properly train and continuously improve an AI engine for offshore route optimization, it is essential to establish a structured feedback loop between the system and its end users. When a result file or proposed routing plan is generated by the AI tool, business expert users such as logistics coordinators and marine planners should be able to review the output and provide structured feedback on its feasibility, accuracy, and alignment with operational realities. This feedback is a critical input for refining an AI model over time. By capturing user insights about what worked, what didn't, and why, the system can learn from real-world decisions and gradually improve its capabilities.

A management SaaS platform can support this process by providing a user-friendly interface for reviewing AI-generated plans and submitting feedback directly within the operational workflow. Over time, this iterative process helps fine-tune the model, making it more accurate, context-aware, and aligned with the unique dynamics of offshore logistics.

Field Testing

The application of an AI-based route optimizer to schedule deliveries of cargo using a vessel fleet has been attempted by Kabal in cooperation with customers using operational data. The scenario included the option to include (at cost) the usage of spot hire vessels and utilizing helicopters to deliver small goods.

Technical Setup

The route optimizer solution was built customized on top of the open source GraphHopper/jsprit framework. This framework addresses routing problems by applying an approach known as *ruin and recreate*. Instead of exhaustively search the solution space of the vehicle routing problem (VRP), which is impossible when a large amount of vessels, platforms, and constraints is included, jsprit instead incrementally improves possible solutions through random but controlled destruction and reconstruction of solutions. This means that an initial feasible route is generated, and then subsets of vessel assignments are removed either randomly, or by mathematical distance or time-windows. The removed routes are then added back to the schedule, considering cost, time windows, limits, and sailing distances. By iteratively applying this ruin and recreate approach, the algorithm explores multiple possible routes, gradually converging toward near-optimal vessel schedules. For offshore oil and gas scenarios, which is characterized by various constraints, this approach provides a flexible and scalable AI optimization engine.

Optimization Results

The result of live field testing proved reduction of vessel fleet usage by over 40% (removing 5 vessels from a fleet of 12). However, some of the routes suggested seemed illogical to the business end users. The reason for this was that multiple vessels were requested to always load at the supply base in parallel, 24/7 operations were introduced by suggesting base loading at night time (regardless of high base overtime compensation introduced), and vessels often returned early instead of loading backload on normal route schedules. When cost of expensive rental equipment was introduced as a part of the cost function, the optimization engine very often suggested that equipment to be sent on a spot hire vessel or using helicopter transport instead of utilizing existing routes. All these results seems illogical but has a mathematical explanation. However, in order for the result to be acceptable in a real world scenario, the business end users had to adjust some of

the suggested routes and delay loading and offloading times to fit within normal schedules, highlighting the need for continuous human adjustments. After manual adjustments, a saving of 33% of vessels (removing 4 vessels from a fleet of 12) as compared to executed routes was observed, which is a significant saving.

Conclusion

The offshore oil and gas industry is under constant pressure to improve operational efficiency, to reduce costs, and to meet ambitious environmental and safety targets set by management. Among the most promising avenues for transformation is the use of artificial intelligence (AI) to optimize logistics—particularly the routing and utilization of offshore supply vessels (OSVs). These vessels represent a significant portion of operational expenditure, and their movements directly impact fuel consumption, emissions, equipment usage, and personnel safety.

Huge savings have already been proven through better coordination, visibility, and planning across stakeholders.[2] The next frontier lies in leveraging AI to further enhance these efficiencies, particularly through intelligent route optimization and just-in-time delivery models.

A key factor to success is to ensure that all stakeholders communicate their cargo demands days and weeks in advance and collaborate by means of a centralized management SaaS platform, and to ensure that processes are in place to properly gather relevant data, fill in information gaps, be able to provide feedback on results and have a simple approach to implementing AI suggestions into real world operations.

The AI model needs to incorporate the flexibility required by logistics to support the ever-changing operational demands from offshore which impact the vessel schedule as well as recalculating all unplanned cargo requirements.

References

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